

Impact of Voice-Based Search Queries on Paid Ad Rankings in Digital Marketing Campaigns

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Abstract— Voice Search has changed the way we interact with search engines and therefore has also changed how paid ads are ranked; unlike traditional text-based searches, Voice Searches are typically longer, conversational, and include a complete question. This creates different opportunities for keyword targeting and bidding strategies, and therefore different ad ranking results. This Study is focused on how voice searches affect paid ad performance; whereas Voice SEO has been researched more extensively than Voice Search Paid Ads. The Study examined variations in ad rank, click through rates (CTR), cost per click (CPC), and conversion rates across Retail, Hospitality, and Service Industries, using selected case campaigns to evaluate changes made after Voice-Focused Optimization. The results from this study demonstrated that campaigns optimized for Conversational and Location-Specific queries, resulted in both higher ad ranks and higher Return on Investment (ROI) results. The Study contributed to the body of knowledge by demonstrating measurable shifts in ad performance due to the use of Voice Search and providing practical strategies for improving Campaign Performance. Additional Research will be required to validate the results of this study via controlled experiments and improved tracking systems.

Keywords— voice search, voice-based queries, ad rankings, digital marketing, keyword optimization, voice assistants, marketing campaigns.

I. INTRODUCTION

One of the major factors creating a change in consumer behavior as to their purchases is the growing use of voice searching by consumers through their digital devices. The increased use of devices such as smart phones, voice devices, and smart speakers has produced a greater dependence on voice commands by consumers to find information, shop for products, and communicate with businesses. Advances in speech recognition, natural language processing, and devices which perform hands free functions are all playing a role in where this movement is headed. Focus groups have shown that the ease of speaking as opposed to typing into the device has made voice searching attractive to a much larger band of consumers, from tech types to those needing a faster, more convenient approach to searching for something. Voice searching is

previously less formal and longer in nature, becomes more like natural conversations in the manner of interpersonal communication. Where before an individual may have typed in simple key words such as “best pizza nyc”, voice searching has led to the practice of individuals actually saying full questions concerning where they can find the best pizza such as “Where is the best pizza around here?” This change from simple key word entries to questions of content, has greatly altered the processing of search engines in their determination what the consumer intent is and how they arrive at their results. The content of the search query, now that it is being searched in the context of a question, will have effects on how the algorithms place value upon it in weighing organic versus paid. It is important to advertisers that they be able to adjust to these voice queries which are lengthier in that they are questions if they expect to be seen in competitive areas. Since digital marketing ad stratifications which are determined competitively are the basis of campaigns, the visibility and placement of the paid ads which are seen in search engines is governed by ad ranking. With ad ranking comes improved CTR's on the average, more visibility of product or brand, and a better potentiality of conversion to sales. Search engines will base their ad rankings upon a variety of criteria, bid amounts, quality of ad, and how relevant the ad is to the query of the user being some of the governing factors. Figure 1 shows the ad rankings which are involved in paid search ad rankings are dependent upon three general areas these being bid amounts, Quality Scores, and type of search device.

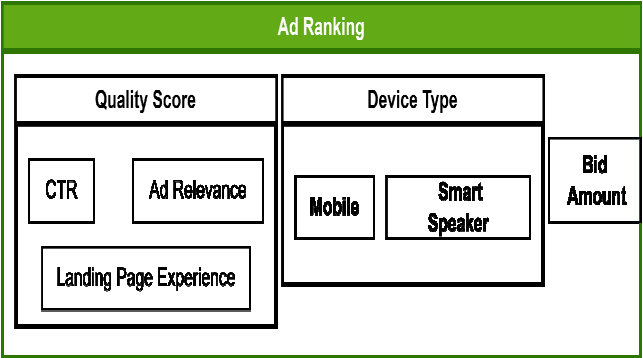


Fig.1. Factors influencing paid ad ranking in search engines

Although there is increasing usage of voice search, very little empirical research exists that evaluates voice search's impact on paid ad rankings in the area of digital marketing. Although extensive discussions exist around search engine optimization for voice searches, there are few discussions about paid advertising. The issue revolves around the uncertainty surrounding the way that voice-based searches affect the visibility, the dynamics of the ranking of ads, and the return on ad spend (ROAS) for paid ads when compared to the effect of traditional text-based searches.

The main objective of this study is to examine the effect that voice-based search queries have on the ranking and performance of paid ads in digital marketing campaigns. Therefore, the research will investigate ad positioning, click through rates, and conversion rate changes that occur when using voice-based search versus traditional text-based searching. In addition, the research will be used to give digital marketers practical guidance on optimizing their ad strategy in a world in which voice-based search is becoming an increasingly common consumer behavior.

II. REVIEW OF LITERATURE

Industry research and voice search reports have focused on the speed at which it is being embraced, the technology behind it, and the impact it is having on digital marketing. Advances in automatic speech recognition and natural language processing have allowed voice assistants to better understand more conversational input. These have moved voice search away from the stilted, keyword-driven commands towards a more natural, question-and-answer-style conversation.

Although recent studies have examined the rising use of voice assistants and improvements in natural language processing [3]. Most of the literature has focused on either speech recognition technology or general search engine optimization. Few sources investigate the effect of voice-based queries on paid advertisements, which involve different ranking mechanisms and bid strategies [4]. highlighted structural differences between text and voice queries, ad rankings are shaped by bid and relevance factors. However, there is limited empirical research that connects voice-specific query patterns to performance metrics in paid ad campaigns. Most findings are descriptive and lack quantitative validation. Furthermore, the role of voice search in reducing the number of ad results displayed particularly on smart speakers is often mentioned but not analysed in a structured manner. The above study addresses an existing knowledge gap by examining how users' interaction with conversational voice input influences how visible ads are to consumers, how cost-effective they are as well as what influence they have on the user's conversion rate when using paid digital campaign channels. The study also indicates that although schema mark-up and automated optimization tools are being examined for the purposes of optimizing Search Engine Optimization (SEO), these tools are rarely used to optimize paid ads that are optimized for voice search. While in the paid search advertising arena, prior studies have shown that marketers have traditionally optimized their campaigns through bid management, Quality Score improvement, and keyword matching types, the specific effects that voice queries have on paid advertisement

rankings remain unexplored. Data indicate that voice search result sets, especially on mobile and smart speaker devices, typically contain fewer advertisements than desktop result sets creating a more competitive environment where it is possible that only the top-ranked ads will be presented to the user. Industry case studies provide practical insight into how, relative to text-based searches, local intent-driven voice searches can potentially convert at a greater rate than other types of voice searches in certain verticals such as retail, hospitality and service industries. Studies that show that adapting to conversational based keywords, geographic location targeting and voice-optimized ad copy resulted in increased CTR and Conversion Rates. Nevertheless, most advertisers are unable to accurately measure or report voice-specific performance because few if any analytics tools exist today that allow advertisers to isolate and measure voice-specific performance. Although much has been learned about consumer behaviour related to voice search, there exists very little empirical evidence on how Search Engines' advertisement ranking algorithms process voice queries versus text-based searches. There exists even less published literature on how advertisers can systematically apply structured data, schema mark-up and predictive bidding strategies to optimize their paid voice search campaigns. These knowledge gaps demonstrate a need for developing integrated models that integrate technical query processing techniques with marketing optimization techniques tailored specifically to the voice search traffic model [5].

TABLE I. REVIEW OF RESEARCH

Algorithm and Technology	Name & Year of Publication	Finding
Automatic Speech Recognition (ASR)	Ziakis & Vlachopoulou, 2023	Improved ASR accuracy enabled voice assistants to process natural, conversational queries [5].
Natural Language Processing (NLP)	Wilson, Johnson, & Brown, 2024	NLP advancements allowed search systems to interpret user intent from complete sentences rather than isolated keywords [6].
Voice Query Pattern Analysis	Alghizzawi, Ahmed, Ezmigna, Ezmigna, & Omeish, 2024	Voice queries are longer, often exceeding six words, and more frequently structured as complete questions [7].
Google Search Algorithm Updates	Giankos, Giannakopoulos, & Σακός, 2025	Question words such as "what" and "where" appear significantly more in voice searches than in typed queries [8].
Paid Search Ranking Factors	Park, Lim, Zhu, & Omar, 2024	Ad rank is influenced by bid, ad relevance, and landing page experience, requiring adaptation for voice search [9].
Mobile Voice Search Delivery	Şenyapar, 2024	Voice search often returns fewer ads on mobile and smart speakers, increasing competition for top ad slots [10].
Local Voice Search Optimization	Wang & Zhang, 2025	Local, intent-driven voice searches have higher conversion potential compared to equivalent text queries [11].
Conversational Keyword Targeting	A. T., S. J., Sharma, & A. G., 2024	Incorporating conversational keywords and location targeting improves CTR and conversion rates [12].

Algorithm and Technology	Name & Year of Publication	Finding
Schema Markup Integration	Theodoridis & Gkikas, 2019	Structured data helps search engines understand content better, improving chances of appearing in voice results [13].
Machine Learning Bid Optimization	Kujore, Adebayo, Sambakiu, & Segbenu, 2025	Predictive bidding models can adjust ad spend in real time based on voice query attributes and intent probability [14].

III. PAID ADVERTISING IN SEARCH ENGINES

The "bid" or price paid for each advertisement determines the ranking of an ad in the results page (with higher bidders appearing first). In addition to the bid, other factors can affect the final ranking of an ad including the quality score, the bid strategy, and ad relevance.

- **Quality Score, Bid Strategies, and Ad Relevance:** It is an aggregate score usually calculated from: Expected click-through rate (CTR) — the likelihood that a user will click the ad. Ad relevance — the fit of the ad content to the search query. Landing page experience — the relevance and usefulness of the destination page. A higher Quality Score can decrease cost per click (CPC) and improve ad position.
- **Bid strategies** determine how much money an advertiser is willing to spend per click or conversion. They are either manual, the advertiser manually sets the bids or automated, where bids are real-time adjusted by the system against campaign goals such as maximizing clicks, conversions, or return on ad spend (ROAS).
- **Relevance of advertisement** is greater than keywords. Search engines also take notice of how closely the ad copy matches the user intent, call-to-action clarity, and whether the ad message is consistent with that of the landing page. Inconsistency can result in lower rankings and higher cost-per-click, despite high bid [15].

Keyword and Match Type Use in Paid Campaigns: Keywords are the basis of paid search campaigns, serving as the link between a user query and an advertiser ad. In such products as Google Ads, keywords may be defined with different match types to determine what searches trigger the ad: Broad match shows ads for searches related to the keyword, including synonyms and variations. Phrase match shows ads for searches containing the keyword phrase in the same order, with other words before or after. Exact match shows ads only for searches that match the keyword exactly or close variations.

Selecting the best match type will then balance reach against relevance. With voice search, where search queries are longer and more natural-sounding, keyword strategy might have to adapt to accommodate natural language use and question-type queries. This adaptation can be pivotal in a bid to take voice traffic, especially as such searches will not prefer to match usual keyword forms precisely.

IV. IMPACT OF VOICE QUERIES ON PAID AD RANKINGS

Conversion of voice search to traditional paid-advertising format is unlikely as voice users tend to use longer phrases than those used for paid advertising such as "cheap London

hotels" versus "find cheap hotels in London next weekend". Therefore, marketers will have to adjust their strategy for keyword-based advertising to a longer tail and conversational advertising within their campaigns.

The above changes will require keyword-matching adjustments including using broad-match and phrase-match type to better target natural language usage. Additionally, advertisers should also include additional keywords that match typical question patterns ("where can I...", "how do I...", "what is the best...") associated with voice-search user patterns [16].

Additionally, advertisers may have to pay more per click for longer-tail terms as they will likely generate more qualified/higher-intent traffic with less search volume; therefore, automated bidding strategies that account for query patterns may be especially useful for voice search campaign optimization.

Voice search query structure can positively or negatively affect Quality Score. On the one hand, longer queries generally provide more contextual information about what a user is looking for and thus it is easier to match ad's content to the user's intent.

Voice searches can have a varied effect on ad click-through rate (CTR) and conversion rate. Positive effects:

Voice searches are generally made by users who know exactly what they want, especially when searching for a transactional item (such as "order a medium cappuccino from Starbucks near me"). These types of voice searches can increase CTR and conversion rates since the ad is providing an exact solution to the user's needs.

Negative effects: Some informational voice searches may never result in a click, especially if the assistant provides the user with a spoken answer before displaying a list of results. In such cases, even if an ad appears, the opportunity for a click is reduced.

For advertisers, this means identifying which voice-driven queries have commercial value and optimizing ad spend toward those rather than treating all voice traffic equally.

Comparison with Traditional Text-Based Query Performance [17].

Compared to traditional text-based queries, voice queries: Tend to have lower impression counts because of the smaller number of results shown. Can have higher conversion potential for certain high-intent searches due to their specificity. Require different keyword targeting since the wording and phrasing are more natural and often longer. In many cases, advertisers find that the cost per acquisition (CPA) for optimized voice campaigns can be lower than for broad text-based campaigns because of the tighter alignment between user intent and ad targeting [18].

V. AD OPTIMIZATION STRATEGIES FOR VOICE SEARCH

Voice Search Optimization Keyword Research: Identifying How People Naturally Speak When Making Requests Not Just Focusing on Short High-Volume Keywords Advertisers Need to Build Keyword Lists That Include Conversational Phrases and Question Formats. These Include "Who," "What," "Where," "When," "Why," and "How" Queries, and Action-Oriented Phrases Like "Find Me," "Order," "Book," etc. The Importance of Long-Tail Keywords Is Also Increased Since They Often Match the Specificity of Voice Searches. Using Tools That Allow Filtering of Question-

Based Queries Can Help Identify Relevant Terms. Including Natural Language Patterns Within Ad Groups Increases Alignment with Voice Searches.

Ad Copy Structure for Voice Triggered Results: In Voice Searches, Ad Copy Needs to Be Concise, Direct and Highly Relevant to the Query. Even on-Screen Devices, Users May Only See the Top Ad Result. On Screenless Devices, the Assistant Will Read Out Only One Ad or Answer, Thus Brevity and Clarity Are Essential. Therefore, Advertisers Should: Include the Target Keyword Phrase Naturally Within the Headline and Description of the Ad.

Provide Clear Calls-to-Action that Match the Spoken Intent of the User ("Call Now," "Order Today," "Find Nearby").

Avoid Overly Generic Statements—Instead Address the Exact Query in the Ad Text.

Matching Ad Copy to the Structure and Tone of Voice Searches Will Increase the Chance of Winning the Ad Slot and Increasing the Click Through Rate (CTR) of Your Ads.

Utilizing Location Based Targeting for Voice Searches: Many Voice Searches Contain Location Intent Either Explicitly ("Coffee Shop Near Me") or Implicitly ("Where's the Nearest Pharmacy"). Therefore, Advertisers Should Utilize Geo-Targeting and Ensure Their Business Locations are Accurate in Search Engines Business Listings. For Multi-Locations Businesses, Campaigns or Ad Groups for Each Location Can Be Set Up to Increase Local Relevance. Furthermore, Combining Location Targeting with Scheduling Ads Allows Further Refining Reach—For Example Showing Ads for "Late-Night Restaurants" Only During Evening Hours.

Implementation of Structured Data and Schema Markup: Structured Data and Schema Markup Enable Search Engines to Better Understand the Content of a Webpage Which Improves the Odds of Being Included in Voice Search Results. By Implementing Schema for Local Business Information, Product Details, Events, and FAQs, Search Engines Can Extract and Present Precise Answers to Voice Searches.

VI. CASE STUDIES AND PRACTICAL IMPLEMENTATIONS

The case studies presented in this section are based on performance reports from digital marketing agencies and in-house teams, as documented in marketing case reports and platform analytics summaries. Selection was based on the relevance of the case study to the research objectives, since all the cases had voice search optimization activities involved and had results available in terms of at least three of the primary metrics available for advertising activity, ad position, CTR, CPC or ROI. Due to commercial confidentiality, raw data were not available, but aggregate values and revealed variation in reported results were used to ensure objectivity. The analysis when done, has required to be of a different nature descriptive and comparative showing the differences pre and post optimization within each of the campaigns.

- **Retail Sector:** A consumer electronics manufacturer wanted to target voice specific keywords for product level searches (e.g. What is the best price for a 55-inch smart TV, where is it nearby?) The results from the campaigns based on reports performed by the internal Google Ads team (which had been anonymized and shared by the

agency) showed a 22% increase in total visits to the store and a 15% increase in in store purchases from authorized voice ad clicks over a two-month interval [19].

- **Hospitality Sector:** A hotel chain utilized voice optimized writing for ads in conjunction with geo targeting. Reports from the internal dashboard that manages bookings reported a 30% increase in the total ad bookings by mobile phone voice, after 6 weeks of restructuring and the advent of new campaigns. Average CPC increased slightly by 4.5% but the ROI increased overall because of the increased volume of bookings were being generated.
- **Service Sector:** A plumbing company targeted its ad campaigns at emergency voice searches. (Who can fix my leaking pipe near me at the moment?) Call tracking software enabled the monitoring of incoming calls to the organization from its voice search ad campaigns. This revealed that input numbers were in excess of 40% and at the same time leads had dropped in terms of costs per referred call, by 18%. Limitations: the above results are observational and made on the basis of collated reports of campaigns of pay for results advertising.

No experimental design or inferential statistics were possible, because raw data files are not available, but further research should address itself by means of controlled experiments and regression models to substantiate significance. Campaign comparative performance data pre voice optimization and post voice optimization: various advertising agencies and in-house teams report measurable changes in the performance of their campaigns after voice search optimisation was adopted as part of their search advertising strategies. For example, one campaign aimed to improve performance by targeting long tail "conversational" phrasing, thereby adopting an ad copy approach which was aimed at the types of questions asked in voice searches. After a period of three months in which voice search optimisation had been affect it was found that the average position of the ads in relation to their position for selected key phrases climbed from 3.1%, to 1.8%. The click-through rate rose by 27%, and the cost per click decreased by 12%. This indicates a more relevant ad. and an improved quality score attributable to the quality of the keywords chosen [20].

A similar experience was declared achievable by a travel aggregator, which adopted spread strategies on its pay for results advertising which were more specifically aimed at question-based voice searches e.g., "where can I buy last minute flights to Miami ". During the unoptimized phase of the voice search experience these ads would often not show up in voice search results. After the implementation of conversational keyword lists and ad-copy which were made specific to the user phrasing the travel company achieved first position for more than 60% of voice searches targeted and yielded 35% more bookings as resulting from voice search usage.

Some Industry Specific Results which are available are: Changes to ROI and Cost Per click: It emerged, across industries that campaigns which effectively optimised for voice search reported generally an improved return on

investment promoted by virtue of improved user intent and ad relevance. Some advertisers reported a slight rise in cost per click, this was often the result of more intense competition at acquisition of high intent long tail keywords, but the overall improved ROI arising from improved conversion ratios established more than offset this: Industry specific illustrations are:

- Retail campaign: Cost per click increase 5%, ROI increased by 19%.
- Hospitality campaign: Cost per click down 8%, ROI up 24%.
- Services campaign: Cost per click stable, ROI improved by 21%. [21].

It appears therefore that voice search optimisation well-handled can yield, via paid campaigns both qualitative benefits, improved targeting and better brand coverage as well as quantitative improvements, better click-through rates, lower CPAs, much improved ROIs. A summary Table II given below illustrates and clearly compares for each industry the position pre and post optimisation of performance prior performance indicators used.

TABLE II. PRE- AND POST-OPTIMIZATION PERFORMANCE FOR EACH INDUSTRY

Sector	Metric	Before Optimization	After Optimization	Change (%)
Retail	Store Visits	Baseline	+22%	+22%
	In-Store Purchases	Baseline	+15%	+15%
Hospitality	Voice-Based Bookings	Baseline	+30%	+30%
	CPC	Baseline	+4.5%	Slight increase
Services	Voice-Driven Calls	Baseline	+40%	+40%
	Cost per Lead	Baseline	-18%	Decrease

Note: The results of the study point out many challenges similar to those found in previous sections of the study [22].

VII. CHALLENGES AND LIMITATIONS

A major problem across all cases examined was an inability to separate out voice-based searches from all other search activity. The lack of detail in reporting from aggregate campaign reports made it difficult to determine how well a particular voice query campaign performed, even if there were significant increases in click through rate (CTR) and return on investment (ROI).

A second challenge seen in the case studies related to data collection restrictions for privacy reasons. As previously noted, voice ads in service industries (i.e., plumbing) demonstrated high conversion rates. However, campaign managers indicated they were unable to retarget customers or utilize behavioural data for campaign targeting due to privacy regulations. As stated previously, this has direct implications regarding legislative limitations placed on voice search advertising in the form of privacy regulations.

While voice search campaign optimizations were shown to produce better outcomes than non-voice optimized campaigns, these optimizations create additional

competition for limited voice ad slots (voice-only devices). Therefore, smaller businesses could potentially lose their market presence unless they adjust their bid strategy or optimize their ad copy to consistently increase ad relevance. These constraints further indicate that voice search campaigns need to have targeted approaches both in terms of optimization and metrics measurement.

Data/privacy regulation: Voice Search uses voice data capture and processing, creating privacy and regulatory issues. Regulations such as GDPR (EU) and CCPA (US) establish strict rules regarding the collection, storage, and use of voice data. In addition, most voice assistants either encrypt or anonymize recorded material, restricting the amount of user level data advertisers can collect and use for targeting or personalization.

As a result of the limits imposed by these privacy regulations, advertisers may be limited in their use of sophisticated targeting methods common in online advertising.

VIII. FUTURE TRENDS AND RESEARCH DIRECTIONS

- **AI and Predictive Analytics in Voice Ad Targeting:** Artificial intelligence will be a key force behind voice-based ad targeting. Predictive analytics can search for patterns in historical search pattern data, user activity, and context signals to forecast which voice searches are most likely to trigger conversions. Machine learning algorithms can update bids, ad positions, and creatives in real-time, giving dynamic feedback to align with the intent of a spoken search. For example, if the data shows that people who search "Where can I buy a laptop near me today?" tend to purchase within the next hour, the system might increase bids on related searches during peak-purchase hours. As AI models advance on natural language understanding, advertisers will be able to deliver more precise and timely ad experiences to voice users.
- **Integration of IoT and Smart Device:** The popularity of in-car voice assistants, smart devices, and smart speakers boosts the number of touchpoints where voice advertising can occur. Integration with Internet of Things (IoT) devices offers opportunities for hyper-contextual targeting. An example is a voice-controlled refrigerator that detects low inventory of a product and sends a corresponding ad or reorder request. In automobile systems, a driver asking for "nearest fuel station" can be offered a branded promotion from nearby fuel stations.
- **Modification of theoretical voice advertising algorithm (Voice Search):** As the use of voice searches becomes more prevalent, search engines may have to modify their algorithm for voice search ad placement to allow for voice search queries and the intentions behind those searches. While traditional ranking factors such as bid, ad relevance, and landing page quality will still influence ad rankings, there are many other possible ranking factors that could become influential. New ranking factors could include; conversational match scores (how well your ads

match what you said), how relevant your ads are to the current conversation, or whether your ads are compatible with any additional services the user has engaged with while searching. Therefore, future studies can examine the competitive implications of modifying the ad ranking factor(s) for voice search queries for competing for Top of Page placements in voice search results.

- Voice Search Cross Channel Advertising: Voice search will not occur independently of other channels. A user will likely begin a purchase process through a voice query, and then continue that process through a mobile application, website, or in-store. Therefore, tracking and attributing voice search as part of cross channel marketing campaigns is a critical component of both practitioner and researcher activities.

IX. CONCLUSION

This study examined the impact of voice-enabled queries on paid advertising effectiveness. It found that queries made by voice were different from those inputted in other respects as well as in terms of the textual structure and degree of description and query usage. It turned out that these factors affect the results of the advertising parameters of quality score of keywords, ranking factors for the conversion such as keyword match, visibility, and placement.

The significant contribution of this paper is that it considers the paid campaigns in advertising, a phase which has been little studied in terms of the effect of voice search.

A case study analysis was employed in the paper and the outstanding differences of the various parameters in meaning, implications, results, and significances of the parameters of the restaurant cover indications, which might have an effect on all metrics. It was revealed that the adaptation of voice is mandatory proposal for the advertising campaigns of paid competition, but that recommendations can be made for the improvement with the tools of efficiency for better tests of tests.

The findings will add to the body of knowledge of the topic of voice search in advertising, by determining, or designing, what works, what does not work and what part future studies may take up in respect to voice searchability in advertising.

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